NYC Taxi Demand Forecasting Project Summary

This project develops a predictive forecasting system for NYC taxi demand using advanced machine learning and deep learning models (SARIMA, Random Forest, LSTM) to transform taxi operations from reactive to proactive fleet management. By accurately predicting passenger demand 30 minutes to 24 hours in advance, the system enables dynamic driver positioning and resource allocation, with the potential to reduce average passenger wait times by 15-25% and decrease operational costs through optimized fleet utilization. The implications extend beyond operational efficiency to enhancing customer satisfaction, reducing urban congestion through better demand-supply matching, and establishing a data-driven foundation for smart city transportation systems that can adapt in real-time to changing demand patterns and improve overall urban mobility infrastructure.

Key Technologies: Python, TensorFlow/Keras, Scikit-learn, Pandas, Time Series Analysis
Models Evaluated: Naive Forecasting, SARIMA, Random Forest, LSTM Neural Networks
Dataset: NYC Taxi Trip Counts (Hourly frequency, ~10,320 data points)
Best Performance: LSTM model with MAE of ~6,151 trips